



Alberta Public Health Disease Management Guidelines

Botulism

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Revisions

Revision Date	Document Section	Description of Revision
August 2023	General	• Entire guideline updated to align with current information • Revision history table moved to front of guideline • Appendix 2 removed as information can be accessed in the Alberta Immunization Policy and product monograph

Case Definition

Confirmed Case

Foodborne Botulism

Clinical illness ^(A) with laboratory confirmation of botulism intoxication:

- Detection of botulinum toxin in serum, stool, gastric aspirate or food,^(B)

OR

- Isolation of *Clostridium (C.) botulinum* from stool or gastric aspirate;

OR

Clinical illness^(A) and indication that the client ate the same suspect food^(B) as an individual with laboratory confirmed botulism intoxication.

Wound Botulism

Presence of a recent wound infection in the two weeks before symptom onset with laboratory confirmation of intoxication and no evidence of consumption of food contaminated with *C. botulinum*:

- Detection of botulinum toxin in serum,

OR

- Isolation of *C. botulinum* from a wound.

Infant Botulism

Laboratory confirmation of botulism intoxication and symptoms compatible with botulism in a person less than one year of age^(C):

- Detection of botulinum toxin in stool or serum,

OR

- Isolation of *C. botulinum* from the patient's stool, or at autopsy.

Colonization Botulism (Intestinal Toxemia)

Laboratory confirmation of botulism intoxication with clinical illness^(A) in a patient aged $\geq$ one year, with severely compromised gastrointestinal tract functioning (i.e., abnormal bowel) due to various diseases such as colitis, intestinal bypass procedures or in association with other conditions that may create local or widespread disruption in the normal intestinal flora.

- Detection of botulinum toxin in stool or serum,

OR

- Isolation of *C. botulinum* from the patient's stool, or during an autopsy.

^(A) Clinical illness is characterized by diplopia, blurred vision, and bulbar weakness. Symmetric paralysis may progress rapidly. See [Table 1: Clinical Presentation](#) for more information.

^(B) Foodborne – identification of organisms in a suspected food is helpful but not diagnostic because botulinum spores are ubiquitous. Therefore, the presence of toxin in a suspected contaminated food source is more significant.

^(C) Clinical illness in infants is characterized by constipation, loss of appetite, weakness, altered cry and loss of head control.

Probable Case

Foodborne

Clinical illness^(A) and consumption of a suspect food item^(D) within the incubation period (12–48 hours).

Suspect Case

Clinical illness^(A) in a person without laboratory confirmed botulism intoxication or an epidemiological link (e.g., exposed to the same source such as food)

^(A) Clinical illness is characterized by diplopia, blurred vision, and bulbar weakness. Symmetric paralysis may progress rapidly. See [Table 1: Clinical Presentation](#) for more information.

^(D) Foodborne – identification of organisms in a suspected food is helpful but not diagnostic because botulinum spores are ubiquitous. Therefore, the presence of toxin in a suspected contaminated food source is more significant.

Reporting Requirements

Physicians, Health Practitioners and Others

- Physicians, health practitioners and others shall notify the Medical Officer of Health (MOH) (or designate) of the zone, of all confirmed, probable, and suspect cases in the prescribed form by the Fastest Means Possible (FMP).

Laboratories

- All laboratories shall report all positive laboratory results:
 - by FMP to the MOH (or designate) of the zone, and
 - by mail, fax or electronic transfer within 48 hours (two business days) to the Chief Medical Officer of Health (CMOH) (or designate).

Alberta Health Services and First Nations and Inuit Health Branch

- The MOH (or designate) of the zone where the case currently resides shall notify the CMOH (or designate) by FMP of all confirmed, probable, and suspect cases.
- The MOH (or designate) of the zone where the case currently resides shall forward the initial Notifiable Disease Report (NDR) of all confirmed and probable cases to the CMOH (or designate) within one week of notification and the final NDR (amendment) within two weeks of notification.
- For out-of-province and out-of-country reports, the following information should be forwarded to the CMOH (or designate) by phone, fax or electronic transfer by FMP including:
 - name,
 - date of birth,
 - out-of-province health care number,
 - out-of-province address and phone number,
 - positive laboratory report (cases), and
 - other relevant clinical/epidemiological information.

Additional Reporting Requirements

- Canadian Food Inspection Agency (CFIA)
 - In the event of a confirmed, probable or suspect case of botulism related to a food product, the MOH (or designate) should immediately notify the Canadian Food Inspection Agency (CFIA) [Alberta Regional Recall Coordinator](#) at 1-587-230-2518.
 - Mon – Fri: from 8:00am to 8:00pm MT,
 - Weekends and statutory holidays: from 10:00 am to 6:00 pm MT.

Clinical Assessment and Epidemiology

Etiology

Botulism is caused most commonly by *Clostridium botulinum*, and rarely by *C. baratii*, *C. butyricum* and *C. argentinense*. These are gram-positive, spore forming bacteria that produce a lethal neurotoxin, botulinum toxin. The spores formed from the bacteria are commonly found in the environment but only grow and produce toxins in the following conditions: anaerobic, temperatures between 3 and 37 degrees Celsius, low acidity, moisture, sugar, and salt. There are seven serotypes (types A–G); serotypes A, B, E and F can cause illness in humans.^(1–3)

Clinical Presentation

The botulinum neurotoxin (BoNT) blocks nerve functions causing paralysis. Onset is typically with neurological symptoms involving the cranial nerves causing symptoms such as:

- difficulty in speaking (i.e., stuttering/stammering),
- dry mouth, double vision,
- drooping eyelids (ptosis),
- facial weakness,
- difficulty swallowing, and
- absence of fever.

These symptoms progress as a descending symmetric paralysis starting with weakness in the neck and arms progressing to other extremities and respiratory muscles. Mechanical ventilation may be required to manage severe respiratory problems. Symptoms may be sudden or progress gradually over several days.^(1,4–7) The overall case fatality rate of botulism can be high if untreated with antitoxin and supportive care. Administration of antitoxin neutralizes the toxin and stops toxin uptake and should be given within 24–48 hours of onset of symptoms for maximum benefit. However, antitoxin should be administered regardless of time since symptom onset to prevent further toxin uptake.^(1,8) For clinical symptoms of various types of botulism, refer to [Table 1: Clinical Presentation](#).

Table 1: Clinical Presentation^(2,3,10,11)

Types of Botulism	Sources of Infection	Symptoms
Foodborne	Ingestion of preformed neurotoxin in food or drink	- Starts with vomiting and/or diarrhea, followed by one or more of: - ptosis (drooping eyelids), - visual disturbance, - fixed and dilated pupils, - dysphagia (difficulty in swallowing), - dry mouth and dysphonia (difficulty speaking). - Symptoms may extend to a descending symmetrical flaccid paralysis - Case is alert and afebrile - Constipation is common later in presentation. - Case-fatality rate is approximately 5%
Wound	Contamination of wounds by: - soil or gravel - injection of illicit intravenous drugs especially black tar heroin	- Inflammation around the wound, followed by low blood pressure and circulatory collapse - Similar symptoms as foodborne botulism - NO vomiting and/or diarrhea - Case-fatality rate is approximately 10%
Infant	Ingestion of spores that germinate in the intestines of infants and produce bacteria that release toxin.	- Symptoms start with constipation and may include: - loss of appetite, - generalized weakness, - weak cry & suck, - droopy eyelids - sluggishly reactive pupils, - blunted facial expression, - disconjugate gaze - drooling, - decreased anal sphincter tone, - hypotonia, - a significant loss of head control. - Case-fatality rate is less than 1%
Colonization (Intestinal Toxemia) • Rare; not well understood	Germination of <i>C. botulinum</i> and production of toxin in the digestive system.	• Similar to foodborne
Inhalation • Rare; does not occur naturally	Accidental or intentional exposure to aerosolized botulinum neurotoxin	• Similar to foodborne
Iatrogenic • Rare; does not occur naturally	High-concentration botulinum toxin A or B for cosmetic or therapeutic purposes	• Similar to foodborne

Diagnosis

Diagnosis is based on high clinical suspicion and a thorough neurological assessment. Myasthenia gravis, stroke and Guillain-Barré syndrome are part of the differential diagnosis and should be ruled out in order to provide antitoxin treatment as soon as possible. Laboratory confirmation of botulism intoxication can be obtained with serum and stool assays for BoNT, stool microscopy for spores, stool cultures, and wound cultures in the case of wound botulism.^(2,11,13) Consultation with the Microbiologist on call at the Public Health Laboratories (ProvLab) is required prior to specimen collection. Testing of suspect cases should be coordinated through the MOH (or designate). Specimens collected (food and clinical) are sent by ProvLab to the Botulism Reference Laboratory in Ottawa, Ontario. For more information refer to the [ProvLab Guide to Services](#).

Treatment

Treatment of suspected botulism cases should be initiated as soon as possible and consists of immediate hospital admission, antitoxin administration, respiratory support, and close monitoring, as required. Wound botulism requires debridement of the infected area and antibiotics.⁽¹¹⁾

Botulism antitoxin (BAT) is used for the treatment of adults, pediatric and infant cases with suspected, probable and confirmed botulism.

- Consultation with an infectious disease specialist should be done prior to the administration of BAT. Botulism Antitoxin (BAT) use must be approved and accessed via the Office of the Chief Medical Officer of Health (OCMOH).
 - See [Appendix 1: Process for the Approval and Release of Botulism Antitoxin](#)
- Refer to the product monograph and the [Alberta Immunization Policy \(AIP\)](#) for detailed instructions regarding access, dosage and administration for the product supplied.
- **NOTE:** BabyBIG can be used to treat infant botulism in children under the age of one. However, since BAT has been authorized for treatment of botulism in all ages and is readily available in the province, this will be the only product in use in Alberta.

Reservoir

Botulism spores are ubiquitous in soil worldwide. They are frequently recovered from agricultural products, including honey. Spores are also found in marine sediments and in the intestinal tract of animals and fish.^(11,13)

Transmission and Incubation Period

For information on transmission and incubation periods for the different types of botulism, refer to [Table 2: Botulism Transmission and Incubation periods](#).

NOTE: Botulism toxin is considered one of the most potent organisms and has the potential for use as a bioweapon. ⁽¹⁻³⁾

Table 2: Botulism Transmission and Incubation Periods ^(1,2,7,10–13)

Types of Botulism	Transmission	Incubation Period
Foodborne	• Ingestion of contaminated food/drink when toxin is produced due to the following: - improperly processed, canned, low-acid, or alkaline foods, - in pasteurized and lightly cured foods held without refrigeration	• 12 – 36 hours (range is six hours to 10 days) after toxin ingestion.
Wound	• Occurs when a wound becomes infected with <i>Clostridium botulinum</i> and toxin is produced.	• Four to 14 days. Average 10 days
Infant	• Ingestion of botulinum spores that germinate in the intestinal tract and produce toxin • Transmission occurs in infants less than one year of age	• Three to 30 days from the time of exposure to the spore-containing substance
Colonization (Intestinal Toxemia)	• Occurs when <i>Clostridium botulinum</i> germinates and produces toxin in the digestive system.	• Unknown
Inhalation	• Transmission occurs when toxins become aerosolized either accidentally or when intentionally used as a bioweapon • Aerosolized toxins do not occur in nature • Has been documented in laboratory workers	• Not well defined. • Estimated at 12-80 hours. • Can be longer if levels of intoxication are lower
Iatrogenic	• Accidental injection of the botulism neurotoxin into the systemic circulation	• Unknown

NOTE: Generally, the shorter the incubation period, the more severe the disease and the higher the case fatality rate.

Period of Communicability

Botulism is not transmitted person-to-person.^(1,2)

Host Susceptibility

Susceptibility is universal. Infection with botulism does not confer immunity from future exposures.⁽¹⁾

Infant botulism affects children younger than one year of age.

- Anyone who eats food contaminated with botulism toxin is at risk of foodborne botulism.
- The risk for wound botulism is higher in people who inject drugs, especially the use of black tar-heroin.
- The risk for iatrogenic botulism is high in those receiving high doses of botulinum toxin for therapeutic treatment.^(2,13,14)
- Although rare, adults with a history of bowel surgery or use of antibiotics may be at increased risk for colonization (intestinal toxemia) botulism.

Incidence

Botulism cases are rare; however, outbreaks of foodborne botulism due to improper food preparation and preservation have been reported globally.⁽²⁾

In Canada, cases of botulism have been nationally notifiable since 1933. The following sources of foodborne botulism have been identified in Canada: fermented fish/marine mammal meat, improperly stored low acid fruit juices and improperly home-canned low acid foods. Raw (unpasteurized) honey is a risk for infant botulism. An average of six cases were reported annually in Canada from 2010-2020.⁽¹⁵⁾ For current case counts in Canada refer to the [Notifiable Diseases Online](#) website.

In Alberta, botulism cases have been reported since 1962 and for the past five years, an average of one case has been reported every year. For more information on reported cases in Alberta, refer to the [Interactive Health Data Application \(IHDA\)](#).

Public Health Management

Key Investigation

- Confirm that the individual meets the case definition.
- Obtain a history of illness including the date of onset and signs and symptoms.
- Determine the possible source of the infection taking into consideration the incubation period, reservoir, and mode of transmission.
- Determine the type of botulism i.e., infant, wound, foodborne, colonization (intestinal toxemia), inhalation or iatrogenic and refer to [Table 3: Other Key Investigation Actions based on Botulism Type](#)
- Contact the Microbiologist-On-Call at the ProvLab for information on collection and transportation of clinical specimens
NOTE: testing and obtaining lab confirmation should not delay administration of antitoxin. Refer to the [Treatment](#) section.

Table 3: Other Key Investigation Actions based on Botulism Type

Types of Botulism	Key Investigation Actions
Foodborne	• Take a detailed food history including recent consumption of potentially contaminated food, home canned food, or any suspected commercial food source • Identify others who may have been exposed to the same source • Consult the Microbiologist-On-Call at the ProvLab regarding collection of food samples
Wound	• Determine the following: - any wounds or injuries in the past two weeks - if wound was contaminated by gravel or soil - any history of injection drug use
Infant	• Take a detailed food history including recent consumption of potentially contaminated food, including any ingestion of honey • Consult the Microbiologist-On-Call at the ProvLab regarding collection of food samples • Identify others who may have been exposed to the same source
Colonization (Intestinal Toxemia)	• Determine any underlying medical conditions that may affect the gastrointestinal system or normal patient flora e.g., colitis, any treatment with antibiotics, or surgery
Inhalation	• Determine occupation e.g., laboratory worker • Identify others who may have been exposed to the same source • Consider the potential for bioterrorism exposure
Iatrogenic	• Determine history of therapeutic or cosmetic injections with botulinum toxin

Management of a Case

- Requires immediate emergency medical treatment. See [Treatment](#) section
 - Treatment must not wait for laboratory confirmation.
- Hospitalized cases should be managed in accordance with hospital infection prevention and control guidelines.
- Provide general education regarding disease transmission and prevention.

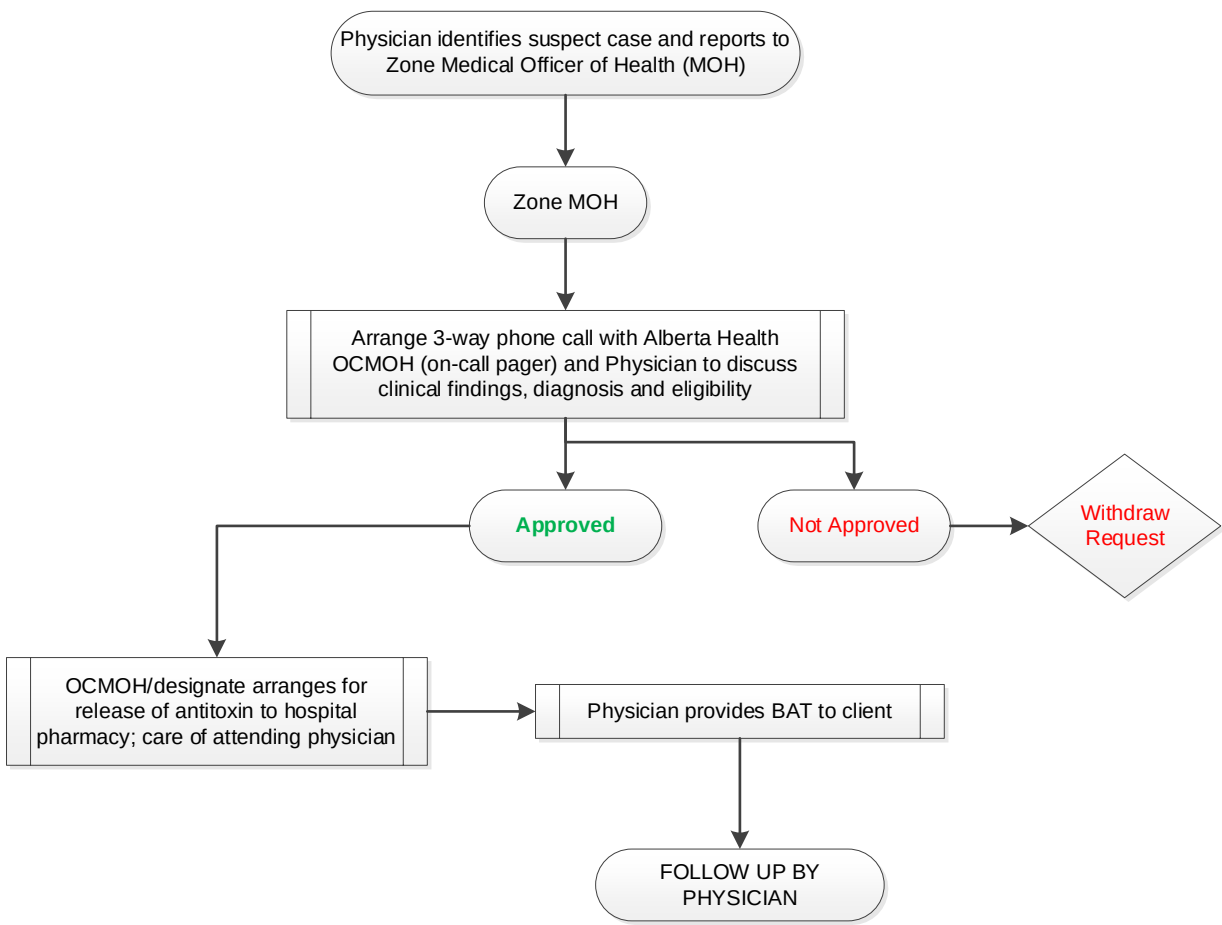
Management of Contacts

- Botulism is not transmitted person-to-person; therefore, contact follow-up is not required.
- However, individuals exposed to the same source (e.g., consumed same food source; same aerosolization exposure) should be assessed for clinical illness and/or advised to seek immediate medical assessment if symptoms develop.

Preventative Measures

- Educate the public on safe food handling as outlined on the [food safety tips](#) website to prevent foodborne botulism.
- Infants younger than one year of age should not be fed honey to prevent the potential consumption of *C. botulinum* spores
- Educate the public on proper wound care to prevent wound botulism. See [Care for a Skin Wound](#) and [Wounds](#) websites for more information.

Appendix 1: Process for the Approval and Release of Botulism Antitoxin



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